Recall

Let W be a subspace of V. We define a relation of V by v_1 is related with v_2 if $v_1 - v_2 \in W$, $\forall\, v_1, v_2 \in V$.

- We show that the above is an equivalence relation on V.
- $v + W = \{ v + w \mid w \in W \}$ is Equivalence class of v.

v_1 & v_2 in V

v_1 is related with v_2

$v_1 - v_2 \in W \Rightarrow v_1 - v_2 = w$ for some $w \in W$

$\Rightarrow v_1 = v_2 + w$

Ex 1 in Tut 2

- V/W = Collection of Equivalence classes of above relation.
- $(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$ $\quad \forall\, v_1, v_2, v \in V$
- $\alpha(v + W) = \alpha v + W$ $\quad \alpha \in F$.

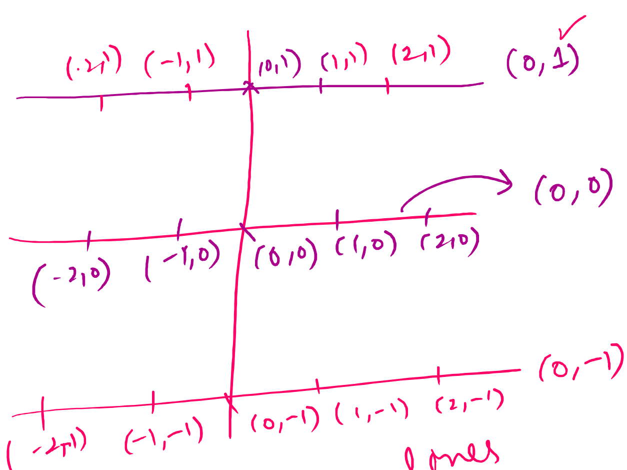
* V/W forms a vector space over F with addition and scalar multiplication as defined above. $0 + W \to$ zero of v.s V/W.

* V/W is called quotient space of V.

- $V = \mathbb{R}^2$, W = x axis

$V/W = \{ (0,y) \mid y \in \mathbb{R} \}$ $\quad (0,y) = (0,y) + x$ axis

(x_1, y_1) & $(x_2, y_2) \in V = \mathbb{R}^2$



(x_1, y_1) is related with (x_2, y_2)

$(x_1, y_1) - (x_2, y_2) \in W = x$ axis

$(x_1 - x_2,\ y_1 - y_2) \in W = x$ axis

iff $y_1 - y_2 = 0 \Rightarrow y_1 = y_2$

Equivalence classes (x_1, y_1)

$= \{ (x_2, y_1) \mid x_2 \in \mathbb{R} \}$

Eq. classes are lines $\parallel$ to $W = x$ axis

$\dim(V) = 2$

$\dim(W) = 1$

$\dim(V/W) = 1$

Eg. classes $\parallel$ to $\underline{W = x\text{-axis}}$ $= \{(x_2, y_1) \mid x_2 \in \mathbb{R}\}$ $\dim(W) = 1$

$(0,1) + W$ is Basis of V/W $\leftarrow$ $\dim(V/W) = 1$

$V/W = \{ (0,y) + x\text{axis} \} = \{ (x,y) \mid x \in \mathbb{R} \}$

$(0,y_1) + x\text{axis} + (0,y_2) + x\text{axis}$
$= (0, y_1 + y_2) + x\text{axis}$

$\alpha\big((0,y_1) + x\text{axis}\big) = (0, \alpha y_1) + x\text{axis}.$

- In vector space $\quad 0 + v = v = 0 + v$

$(0,y) + x\text{axis} + \big((0,0) + x\text{axis}\big) = (0,y) + x\text{axis}$

Basis and Dimension of V/W :-

Let $\{w_1, w_2, -, w_k\}$ be Basis of W $\longrightarrow$ L.I. set, $L.S(w_1, w_2, - w_k) = W$

$\{w_1, w_2, -, w_k\}$ is a L.I. set $\quad W \subseteq V$

$\{w_1, w_2, -, w_k\} \subset V$

- Any L.I. set can be extended to the Basis

$\mathcal{B} = \{w_1, w_2, -, w_k, v_1, v_2, -, v_n\}$ is $\underline{\text{Basis of } V.}$

$\longrightarrow$ L.I. set and $L.S(\mathcal{B}) = V.$

Basis of V/W

$\{\underline{w_1 + W, w_2 + W, -, w_k + W}, v_1 + W + v_2 + W, -, v_n + W\}$

$w_1 + W = W$, $w_2 + W = W$, $w_k + W = W$

$W = \underbrace{0 + W}_{\text{zero in } V/W}$

Claim $\{ \ldots +W\}$ is a Basis of V/W.

Claim $\{v_1+W, v_2+W, \ldots, v_n+W\}$ is a Basis of V/W.

- $\{v_1+W, v_2+W, \ldots, v_n+W\}$ is L.I. set in V/W

$\alpha_1(v_1+W) + \alpha_2(v_2+W) + \ldots + \alpha_n(v_n+W) = 0+W$

$(\alpha_1 v_1+W) + (\alpha_2 v_2+W) + \ldots + (\alpha_n v_n+W) = 0+W$

$(\alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n) + W = 0+W$

$\Rightarrow \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n \in W$.

v_1 is related with v_2

$v_1 - v_2 \in W$ iff

$v_1+W = v_2+W$

- $\{w_1, w_2, \ldots, w_k\}$ is a Basis of W

$\alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n = \beta_1 w_1 + \beta_2 w_2 + \ldots + \beta_k w_k$

$\Rightarrow \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n - \beta_1 w_1 - \beta_2 w_2 - \ldots - \beta_k w_k = 0$

$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \ldots, \alpha_n = 0, -\beta_1 = 0, \ldots, -\beta_k = 0.$

$\because \{v_1, v_2, \ldots, v_n, w_1, w_2, \ldots, w_k\}$ is Basis of V.

- L.S $\{v_1+W, v_2+W, \ldots + v_n+W\} = V/W$.

Let $v+W \in V/W$, $v \in V$

$v = \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n + \alpha_{n+1} w_1 + \alpha_{n+2} w_2 + \ldots + \alpha_{n+k} w_k$

$v+W = \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n + \alpha_{n-1} w_1 + \alpha_{n+2} w_2 + \ldots + \alpha_{n+k} w_k + W$

$\parallel$?

$\alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n + W$

$v_1+W = v_2+W$

$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$

$$v_1 + W = v_2 + W$$
iff $v_1 - v_2 \in W$

$$v + W = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n + W$$
$$= (\alpha_1 v_1 + W) + (\alpha_2 v_2 + W) + \cdots + (\alpha_n v_n + W)$$
$$v + W = \alpha_1 (v_1 + W) + \alpha_2 (v_2 + W) + \cdots + \alpha_n (v_n + W)$$

---

(1) $\{w_1, w_2, \ldots, w_k\}$ is Basis of W

$$\dim(W) = k$$

(2) $\{w_1, w_2, \ldots, w_k, v_1, \ldots, v_n\}$ is Basis of V

$$\dim(V) = n + k$$

(3) $\{v_1 + W, v_2 + W, \ldots, v_n + W\}$ is Basis of V/W

$$\dim(V/W) = n$$

- $$\dim\left(V/W\right) = \dim V - \dim W.$$

Linear Transformations

Let V and W be two vector space over F
A map $T: V \to W$ is called linear transformation if

(1) $T(v_1 + v_2) = T(v_1) + T(v_2)$ $\quad \forall\, \alpha \in F$

(2) $T(\alpha v_1) = \alpha T(v_1)$ $\quad \forall\, v_1, v_2 \in V.$

Example :- $T: \mathbb{R}^2 \to \mathbb{R}^2$

$T(x, y) = (2x, x + y)$

Example :- $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$$T(x, y) = (2x,\ x+y)$$

Check this map T is a linear transformation

T.P. $T\big((x_1, y_1) + (x_2, y_2)\big) = T(x_1, y_1) + T(x_2, y_2)$

$$T\big((x_1, y_1) + (x_2, y_2)\big) = T(x_1 + x_2,\ y_1 + y_2)$$

$$= \big(2(x_1 + x_2),\ (x_1 + x_2) + (y_1 + y_2)\big)$$

$$= \big(2x_1 + 2x_2,\ (x_1 + y_1) + (x_2 + y_2)\big)$$

$$= \big(2x_1,\ (x_1 + y_1)\big) + \big(2x_2,\ x_2 + y_2\big)$$

$$= T(x_1, y_1) + T(x_2, y_2).$$

- $T\big(\alpha(x, y)\big) = \alpha T(x, y)$

$$T\big(\alpha(x, y)\big) = T(\alpha x, \alpha y) = \big(2\alpha x,\ \alpha x + \alpha y\big)$$

$$= \alpha\,(2x,\ x+y)$$

$$= \alpha\, T(x, y).$$

- T is a linear transformation.

Example :- $T : \mathbb{R}^4 \longrightarrow M_2(\mathbb{R})$

$$T\big((\alpha, \beta, \gamma, \delta)\big) = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

Here T is a linear transformation

show that T is a linear transformation.

(1) $T\left[(\alpha_1, \beta_1, \gamma_1, \delta_1) + (\alpha_2, \beta_2, \gamma_2, \delta_2)\right]$

$$= T(\alpha_1+\alpha_2,\ \beta_1+\beta_2,\ \gamma_1+\gamma_2,\ \delta_1+\delta_2)$$

$$= \begin{bmatrix} \alpha_1+\alpha_2 & \beta_1+\beta_2 \\ \gamma_1+\gamma_2 & \delta_1+\delta_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & \delta_1 \end{bmatrix} + \begin{bmatrix} \alpha_2 & \beta_2 \\ \gamma_2 & \delta_2 \end{bmatrix}$$

$$= T(\alpha_1, \beta_1, \gamma_1, \delta_1) + T(\alpha_2, \beta_2, \gamma_2, \delta_2).$$

(2) $T(A(\alpha, \beta, \gamma, \delta)) = A(T(\alpha, \beta, \gamma, \delta))$. Check

Qns :- $T : \mathbb{R} \to \mathbb{R}$ defined by $T(x) = 5x + 3$
check if T is linear transformation. No

$$T(x+y) = 5(x+y)+3 = 5x + 5y + 3$$

$$T(x) + T(y) = (5x+3) + (5y+3)$$

- Any map $T : \mathbb{R} \to \mathbb{R}$ defined by $T(x) = \alpha x + k$ is linear transformation iff $k = 0$.